CS23E01 / EMBEDDED SYSTEMS AND INTERNET OF THINGS

Session: July-Nov 2026 (B.E. CSE – VII sem)

Activities for Lab-2:

1. Perform basic 8-bit basic ALU operations on the 8051 simulator (refer to 'instruction set') using: ADD, SUB, MUL, DIV, ANL, ORL, XRL, RL, RR, CPL, CLR, SETB
2. Perform 16-bit ALU operations using the above instructions and observe the differences.
3. Working with Bit-addressable registers: Access and change a specific bit from the specific RAM location
4. Working with stack memory, performing basic stack operations PUSH, POP
5. Modifying the values of PSW register by programmer.
6. Accessing registers from 80 to FF (SFRs) and print their values.
7. Using instructions for Looping, subroutine Call and unconditional / conditional Jump –

   Loops: Using labels and registers

   Jumps: AJMP, SJMP, LJMP, DJNZ, JC, JNC, JZ, JNZ, JB, JNB, JBC, JC, JNC, JZ, CJNE, JMP

   Calls and returns: LCALL, ACALL, RET, RETI

Observation:

Write your Observations for each of the exercises, for the values of the **PC, DPTR, SP, PSW** and other applicable **SFR**s.

References:

- Edsim51 user guide
- 8051 Instruction set
- Class notes theory pdf